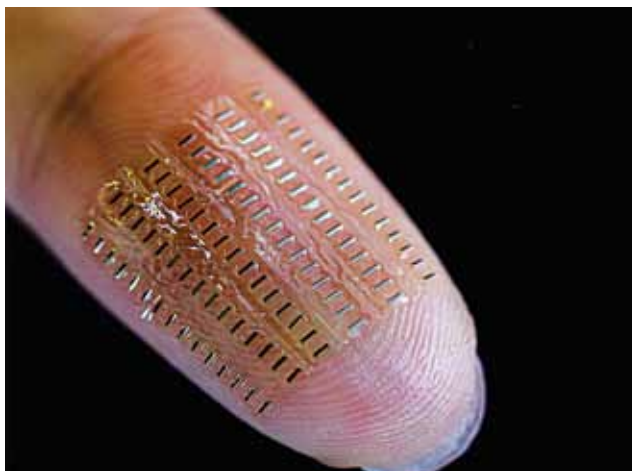


memory. They have developed a phase-change memory cell that operates with low power and can be embedded on flexible plastic substrates commonly used in bendable smartphones, wearable body sensors and other battery-operated mobile electronics. Besides these, phase-change memory could also be used in large data centres where data storage accounts for about 15% of electricity consumption.

A flexible phase-change memory substrate held by tweezers (left) with a diagonal sequence showing substrates in the process of being bent (Credit: <https://energy.stanford.edu>)



Microsupercapacitors to energise intravascular implants and more



The miniaturisation of microelectronic sensor technologies is progressing rapidly. However, it also poses major challenges in the form of energy storage efficiency, ability to operate in small areas, and bio-compatibility with respect to the human body. Now a breakthrough has been achieved by the research team at Chemnitz University of Technology in developing small microsupercapacitors, which can function in (artificial) blood vessels and also be used as an energy source for a tiny sensor system to measure pH. This storage system opens up possibilities for intravascular implants and microrobotic systems for next-generation biomedicine that could operate in hard-to-reach small spaces deep inside the human body.

An array of 90 tubular nano-biosupercapacitors (nBSCs) on the fingertip enable autarkic operation of sensors in blood (Credit: www.tu-chemnitz.de/tu)

Nanosensors for rapid testing of plant hormones

Researchers from the Disruptive & Sustainable Technologies for Agricultural Precision (DiSTAP) Interdisciplinary Research Group (IRG) of Singapore-MIT Alliance for Research and Technology (SMART), and their local collaborators from Temasek Life Sciences Laboratory (TLL) and Nanyang Technological University (NTU), have developed a nanosensor to enable rapid testing of synthetic auxin plant hormones. Current methods cause damage to plants and are unable to provide real-time in-vivo monitoring and information. The novel nanosensors are safer and less tedious than existing techniques for testing plants' response to compounds such as herbicide and can be transformative in

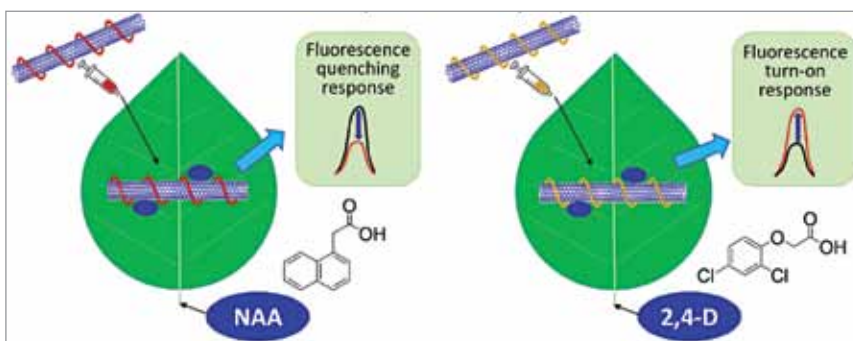


Illustration of novel in planta CoPhMoRe nanosensors for detection of synthetic auxin plant hormones, NAA and 2,4-D (Credit: <https://smart.mit.edu/news-events>)

improving agricultural production and our understanding of plant growth.